

How much should we believe our own disagreement with the market?

Selective predictive betting on European football — correcting the winner's curse at decision time.

REPO · SRP STATUS · SPECIFICATION FROZEN, 0 / 35 BUILT CODE · PIPELINE.PY, 50 STUBS

01 The project in two sentences

We build a football match prediction pipeline, but the research question is not whether our model beats the market — it is **how much of our model's disagreement with the market we should actually believe**, since the bets we select for maximum disagreement are precisely the ones where that disagreement is most inflated by noise. Our contribution is a James–Stein-style **shrinkage of the model's deviation toward the market price before betting**, tested by whether it removes the characteristic decay of the ROI–threshold curve, in simulation and in European league data.

THE MECHANISM

Our estimate is market price plus signal plus noise. We bet only where the gap is largest. That filter selects for large noise as reliably as it selects for large signal — so realised return falls short of predicted return, and the gap widens as we get pickier.

THE GAP IN THE LITERATURE

Hubáček generates disagreement; Walsh & Joshi calibrate probabilities. Verified: nobody *discounts* the disagreement before acting on it. Everyone thresholds on the raw gap. That slot is empty.

WHAT WE DO DIFFERENTLY

Shrink the deviation δ toward the market by a weight w estimated from data, allowed to vary with how noisy the estimate is likely to be. Then threshold. Order matters: calibrate $\rightarrow \delta \rightarrow$ shrink $\rightarrow$ threshold.

Why this is a safe dissertation. The hypotheses concern the **shape** of the ROI–threshold curve, not the level of profit. Market efficiency therefore cannot sink us: if nothing is profitable, the curve shape is still measurable and the result still stands. The alternative framing — "our model beats the closing line" — has no such escape hatch. If the model loses, that project has nothing.

02 Hypotheses

- H1** Realised return declines at high selection thresholds even as predicted expected value rises. The winner's curse operates at decision time.
- H2** Shrinking the deviation before thresholding removes that decline — and does so genuinely, not as a re-parametrisation of the threshold. Must be shown non-vacuous: corrected curve $\neq$ uncorrected curve relabelled in τ .
- H3** The estimated shrinkage weight w is below 1, and varies systematically with proxies for estimate variance — odds level, bookmaker disagreement, book count, division, Elo-vs-ClubElo gap, team history length.

Stated up front, not in limitations: H1 and H2 are characterised in simulation. Football data is *consistent with* the prediction but cannot discriminate at single-season sample size — hence the multi-season design. Declaring our own power ceiling reads as methodological care; leaving it for a reader to find reads as a hole.

03 What the data already tells us

All figures below come from EDA.py on all-euro-data-2025-2026.csv — derived from the data alone, no external source.

7.7%	Bookmaker margin (overround). Raw 1/odds is not a probability — de-vigging is mandatory, not cosmetic.
0.68	Longshot ratio: outcomes priced at 0.113 win 0.076 of the time. The margin is <i>not</i> spread evenly across the odds range, which is why basic normalisation is demonstrably wrong at long odds and Shin/power must be compared.
≈ 1.00	Log loss of the de-vigged market price. This is the number to beat, and the harness sanity check: a dummy model returning the market must score this.
$r \approx 0.0001$	Rolling form against market residual. Expect almost nothing from form features — the market has already priced them.
$\sigma \approx 1.18$	Standard deviation of per-bet P&L → roughly 2,200 bets needed to detect a 5% effect. Drives the multi-season decision directly.
39.3%	Pinnacle coverage — dropped from 23/07/2025 onward, not 58% as first assumed. Closing odds exist only from 2019/20, which fixes the modelling window.

04 Build order

Nine phases, 35 items, sequenced so the things that can kill the project happen early and cheaply. **Phase B is a real gate** — if the ROI decline does not appear in multi-season data, we reframe before writing anything else. Phase E needs no football data and can run in parallel with F.

A DATA

- 1 `fetch_data.py` — adapt §1 of `data_sources.pdf`: widen E0–E3 → 22 divisions, extend seasons back, cache to parquet
- 2 Extend `load_data.py` to multi-season — Pinnacle dropped only from 23/07/2025 (coverage 39.3%, not 58%)
- 3 Pull `xgabora/Club-Football-Match-Data-2000-2025` — ClubElo already merged, team names reconciled. Verify the join on a sample.
- 4 Coverage table per season → fix the modelling window (closing odds exist only from 2019/20)

B GATE

- 5 Re-run the existing `EDA.py` threshold sweep on multi-season data
- 6 **Decision.** Decline holds → continue. Decline flattens → reframe before writing anything else.

C HARNESS

before any modelling

- 7 • Metrics — log loss, RPS, Brier, classwise ECE. Unit-test on synthetic arrays.
- 8 • Walk-forward splitter — chronological, no same-day straddle
- 9 • Bootstrap CIs — resample matches, not bets
- 10 • Gate: run the harness on a dummy model returning the market price. Expect log loss ≈ 1.00 .

D SAFETY

- 11 • Leakage suite — every component changes output; every feature computable at kickoff; shifting features forward one match degrades performance

E SIMULATION

no football data — parallel with F

- 12 • Synthetic p , $\hat{p} = p + \varepsilon$, controlled σ_ε ; sweep τ ; predicted vs realised return
- 13 • Show shrinkage removes the bias
- 14 • Power analysis — bets needed to detect the effect
- 15 • Confirm H2 isn't vacuous — corrected curve $\neq$ uncorrected reparametrised in τ

F PIPELINE

- 16 De-vigging — basic, power, Shin. Pick the winner against observed longshot frequencies (priced 0.113, win 0.076).
- 17 Elo — chronological, pre-match ratings. Record ClubElo alongside.
- 18 Form — 9-match window, four groups. Use the `side == "H"` filter pattern, not positional slicing.
- 19 Rest + congestion — from dates alone
- 20 Bookmaker disagreement, book count, best price
- 21 Models — multinomial logistic with $\text{logit}(q)$ as feature, plus gradient boosting
- 22 Calibration — Platt, isotonic, reliability diagrams. Order: calibrate → δ → shrink → threshold.

G CONTRIBUTION

- 23 Build the σ_ε^2 proxy set — odds level, book disagreement, book count, division, Elo-vs-ClubElo gap, team history length
- 24 • Estimate w on validation → answers H3
- 25 • Apply $\tilde{p} = \sigma(m + w \cdot \delta)$; then heteroscedastic $w(x)$; then Tweedie
- 26 • Threshold sweep, corrected vs uncorrected → **the figure**

H EXTENSIONS

cut from the bottom

- 27 Asian handicap market
- 28 Per-division bias analysis
- 29 Era stratification
- 30 Decorrelation comparison arm
- 31 Experiment runner

I FINISH

- 32 Freeze the specification in writing
- 33 Unlock test set — run once
- 34 Plots
- 35 Write up

Changed since the last version: items 3 and 23 are new — ClubElo enters as a *variance proxy* for $w(x)$, not merely as a bug check, which is the one genuinely useful thing `data_sources.pdf` unlocks. Item 18 now specifies the safe home/away filter pattern. Item 1 adapts existing code rather than starting cold.

05 Reading list

Nineteen papers — enough to show command of the field, few enough to read properly. Grouped by the job each one does.

The betting literature we are contributing to

PAPER	WHAT IT DOES FOR US
Walsh & Joshi (2024) + Corrigendum (2025)	Calibration over accuracy — our primary framework. Verified: uses $\tau = 0$ and never sweeps it. Cite the corrigendum; do not quote the original ROI figures.
Hubáček, Šourek & Železný (2019)	The standard "ML exploits betting market" reference.
Hubáček & Šír (2020)	Our nearest neighbour — and it argues the <i>opposite</i> direction: noise advantages the market taker. We must reconcile with it explicitly.
Constantinou & Fenton — pi-ratings, pi-football	Rating baseline. Also evaluates at "discrepancy levels $\geq 5\%$ " — another instance of our threshold rule in the wild.
Berrar, Lopes & Dubitzky (2019)	Rating features beat recency features; establishes RPS as the metric.

Methodological armour

Winkelmann, Ötting, Deutscher & Makarewicz (2024)	Single-season inefficiencies appear even under full efficiency. Justifies the multi-season design and pre-empts the obvious attack.
Bailey & López de Prado — Deflated Sharpe Ratio	One sentence, defensive: pre-empts "isn't this backtest overfitting?" Distinguish selection <i>among strategies</i> from selection <i>among bets within one strategy</i> .

Winner's curse — provenance

Capen, Clapp & Campbell (1971)	Origin of the winner's curse, in auctions.
Shi et al. (2016), <i>PLoS Genetics</i>	Correction improved performance across all 14 diseases — proof the fix works in practice, not only in theory.
Deng (2019), <i>On Post-Selection Inference in A/B Testing</i>	Directly addresses Open Risk #4. Stronger than a stated caveat.
arXiv 2602.08892 (2026), <i>Winner's Curse Drives False Promises in Data-Driven Decisions</i>	Closest framing match — winner's curse at decision time. Validates the key conceptual move and shows it is current.

The correction itself

Efron (2011), <i>Tweedie's Formula and Selection Bias</i> , JASA	Essential if we implement the Tweedie variant — it is literally our problem, named.
arXiv 2511.06318 — Bayesian hybrid shrinkage (A/B testing)	Bias scales with sampling variability <i>and</i> threshold jointly — the justification for heteroscedastic $w(x)$.

Method details we implement

Shin (1993)	The Shin de-vigging method. Required for item 16.
Štrumbelj (2014), <i>On determining probability forecasts from betting odds</i>	The reference comparison of basic vs power vs Shin. Item 16 is essentially a replication of this.
Snowberg & Wolfers (2010)	Favourite–longshot bias — we measured it (0.68 ratio at long odds); cite the explanation.

Evaluation

Constantinou & Fenton (2012)	RPS for football forecasts.
Zadrozny & Elkan (2002)	Isotonic calibration.

Verification status. Walsh & Joshi and Hubáček & Šír were read in full — those claims are checked. The rest come from the reading list or from search: confirm year, volume and page before submission rather than trusting recall. **Dropped:** the survey (2403.07669) and feature catalogue (2309.14807) — useful while building, not worth citing unless we lean on them; and the GWAS tutorial, which is a teaching resource, not citable — cite Shi et al. instead.

06 Where we are, and what happens next

EDA.py is finished and its outputs are committed. pipeline.py exists as 1,128 lines of **deliberately empty stubs** — 50 functions across sections A–K, each carrying its source paper, its equation in the write-up's notation, and its pseudocode. Nothing is implemented yet. That is by design: the method section can be drafted straight from the docstrings.

Immediate next step — item 1, widening data collection to 22 divisions and multiple seasons, which unblocks the Phase B gate. Phase E (items 12–15) needs no football data at all and should be picked up in parallel by whoever is not on data collection.